**Jeffrey Buskirk | United States Citizen**
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## EDUCATION

**University of Arizona | College of Engineering |** Tucson, AZ
Aerospace Engineering | GPA: 3.84 | Expected Graduation: May 2028 | W.A. Franke Honors College

## SKILLS

**Simulation & Analysis:** CFD (ANSYS Fluent, CFX, ADS, CFD++, STAR-CCM+), meshing (Pointwise, HELDEN), FEA (ANSYS Static Structural, Abaqus), Thermal (ANSYS Transient Thermal, NASA CEA)
**CAD/CAM & Electrical:** NX, SolidWorks, Fusion 360, Onshape and KiCad
**Programming & Hardware:** MATLAB, Python, LabVIEW; Linux/HPC workflows; data automation & post-processing tooling; machining (lathe, mill), additive manufacturing, I²C sensor integration
**AWARDS:** 1st Place Category A – FAR-OUT Competition (Friends of Amateur Rocketry, Oxidizer Uninhibited Tournament), 2× Honeywell Aerospace Additive Manufacturing Contest Winner, Innovation Award (VEX Robotics Worlds), Forklift Certified

## PROFESSIONAL EXPERIENCE

***Turbomachinery Engineering Intern – Boom Supersonic | Summer 2026***

- Owned a mission-critical low-pressure compressor (LPC) stall-margin issue; traced the root cause to high Rotor 1 incidence and ran a design-of-experiments study on IGV and rotor leading-edge angles across 60–100% corrected speed, recovering positive stall margin at off-design conditions without sacrificing 100% speed efficiency.
- Built an automated CFD workflow and AI-integrated dashboard to triage DOE cases by performance impact and rapidly run unsteady CFD on top candidates, including automated speed-line point submission.
- Redesigned the aft-turbine outlet diffuser for high-temperature, high-Mach flow, meeting the <1.77% pressure-loss target while cutting cost ~50% by switching from complicated geometry that would require spin casting to sheet metal.

***Manufacturing and Test Engineering Intern – R3 Aerospace | Winter 2026 – Ongoing***

- Designed and validated high-pressure solid rocket motor bulkheads using ANSYS FEA through hydro-proof and static-fire testing; developed test infrastructure supporting medium-scale SRMs for defense customers.
- Bridged design and production, working directly with manufacturing to resolve fabrication and tolerance challenges leading to successful static-fire and flight testing.

***Chief Engineer, Former Propulsion Manager – Wildcat Rocket Engineering Club | Fall 2025 – Ongoing***

- Led the first team to successfully launch a liquid rocket twice at the FAR-OUT competition, earning 1st Place in Category A.
- Own end-to-end development of a 1,400 lbf nitrous–ethanol liquid engine and student-designed $N_2O$/E98 rocket—from design and manufacturing through custom test infrastructure, 1,500 psi hydro proof, cold-flow, static-fire, and flight testing.
- Lead thermal/fluid design of a 650 lbf regeneratively cooled engine for an invite-only Freeform LPBF challenge; sized 60 coolant channels using Bartz/Dittus-Boelter correlations, validated ~3.6 MW/m² peak heat flux with ANSYS conjugate heat transfer and temperature-dependent 316SS FEA, and used CFD to characterize coolant pressure drop, LPBF roughness effects, and pintle-injector performance.
- Led failure investigation after a hard start on the 1,400 lbf engine's first hot-fire; used ANSYS to identify the combustion-instability driver and redesigned the injector, enabling successful subsequent testing.
- Train and lead a 70-person engineering team across ANSYS, CAD, machining, design reviews, and test operations; secured ANSYS, SendCutSend, and Boltline sponsorships and coordinate technical development across subsystems.

***Engineering Research Associate – The Space Analog for the Moon and Mars | Fall 2025 – Summer 2026***

- Led a 3-person team delivering a space-station $CO_2$-removal prototype for Paragon Space Development and ICES; designed and built an AC electrical control system with manual override and relay automation for adsorption/desorption cycling and led mechanical design for repeated thermal-vacuum operation.

***Turbomachinery Engineering Intern – Honeywell Aerospace (Fans and Compressors) | Summer 2025***

- Analyzed compressor performance maps in ANSYS CFX (post-processed in MATLAB/Excel), improving adiabatic efficiency ~0.5% via clearance and bleed-air studies for Honeywell's next generation turbofan engine.
- Built a Thermodynamic Performance map-to-CFX benchmarking capability new to the team for surface-roughness studies (~1% efficiency gain toward the 10,000 lbf thrust target); ran large-scale CFD on Linux HPC with MATLAB tools that cut post-processing runtime 20%.

***Test Engineering Intern – Honeywell Aerospace | Summer 2024***

- Designed test articles for the updated RS-25 main fuel/oxidizer valves (NASA Artemis SLS) in NX with cryogenic test protocols and ran actuator testing for NASA Gateway docking hardware in cleanroom thermal-cycle environments.
- Programmed (C++), machined, and designed controllers for EVTOL actuation qualification rigs — presented to management and adopted by the customer; also supported TVC actuator testing on a DoD contract.

## PUBLICATIONS

Hentzen, G., Staats, K., Ackerman, A., Buskirk, J., Phoebe, B., & Knox, J. C. "Further Design and Initial AI&T of the Carbon Dioxide Removal System for SAM." 55th International Conference on Environmental Systems (ICES-2026-524), 2026.